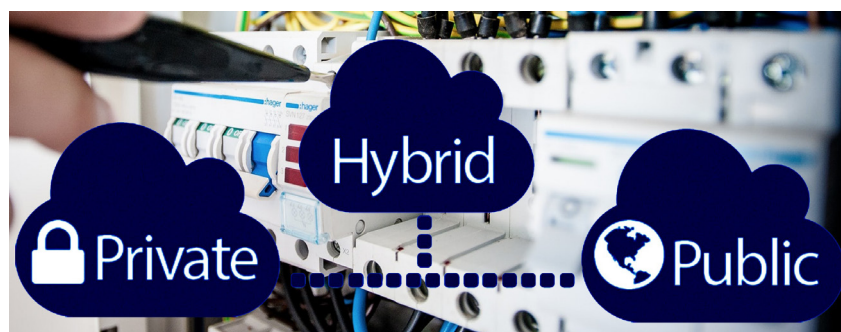


A Complete Guide to Cloud Service Architectures

In its roughly 16 years of evolution, cloud computing has evolved to become a technology that is used by almost everyone who uses the Internet. It can be used as a service to support various types of business and consumer requirements. Therefore, multiple service architectures are being used in cloud computing to customise the technology as per modern day needs. This article provides a complete guide to all the service architectures being used today.



While the idea of having a network of computers collaborating across the world has existed since the early 1960s, the formal conceptualisation of it occurred in 2006 when Eric Schmidt, the then CEO of Google, introduced the term ‘cloud computing’ in its modern day context.

Cloud computing can be simply understood as a network of remote servers across the world, sharing data and collaborating over the Internet to provide services to businesses and customers. Albeit an arbitrary definition, it does cover the core idea behind cloud computing. The primary motivating factor for such a technology was to create more ‘stickability in data’, i.e., to make data more easily accessible across the devices whilst reducing the risks of data loss. If a user x has data in only one server, the chances of permanent

data loss for x are higher given that all it takes is one server outage. That is equivalent to the proverbial ‘putting all your eggs in one basket’ method, which is never a good idea especially when you are dealing with data that can be critical for businesses and consumers. But if you replicate the data of x in multiple servers across the globe, it will have two major benefits. For one, x will still be able to access his/her data even if a server is facing an outage. Second, the cloud can provide x with access to its data from a server that is available closest to it with the least amount of load. This makes data faster and more easily accessible across different devices for x .

In its roughly 16 years of evolution, cloud computing has gone from being something used simply for backing up photos to becoming the backbone of the Internet. Almost every app today,

from Microsoft Office to Asana and Todoist, makes use of cloud computing for real-time access and sharing of data. Almost any app that you can think of uses cloud computing. Everything from Gmail and YouTube, to Instagram and even WhatsApp, uses cloud computing in the background to provide fast, easy, and reliable access to data.

The companies that provide cloud computing services are called cloud service providers. Amazon, Google, Microsoft, Salesforce, Cloud9, etc, all provide cloud as a service in both B2B and B2C contexts.

In the early days, cloud service providers generally offered only three types of services to their customers:

- Software as a Service (SaaS)
- Platform as a Service (PaaS)
- Infrastructure as a Service (IaaS)

However, as the industry requirements have evolved with new technologies such as blockchain and AI coming into the picture, cloud service providers have come up with new models to better serve the varying requirements of their customers. In this article, we are going to go through all the cloud computing service models currently being used in the market.

The architecture of a cloud

Now that we have an idea of what cloud computing is and how it evolved into becoming a 445 billion dollar